

ORF307: Optimization

Spring 2024 1st Midterm Exam Solutions

March 7, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: “I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam.”
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

1. *Problem 1 (35 points). Ridge Regression and physics.*

Let $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$, $\lambda \geq 0$. Consider the following optimization problem:

$$\text{minimize} \quad \|Ax - b\|_2^2 + \lambda \|x\|_2^2. \quad (1)$$

- (a) (5 points) Write problem (1) as a least squares problem. That is, give matrix $\tilde{A}$ and vector $\tilde{b}$ such that (1) can equivalently be written as:

$$\text{minimize} \quad \left\| \tilde{A}x - \tilde{b} \right\|_2^2. \quad (2)$$

- (b) (10 points) Solve problem (2) for the optimal solution x^* . Write your answer to only be in terms of A, b, λ . Which assumptions do you need on A to have a unique solution?
- (c) (10 points) Given wide matrix $C \in \mathbf{R}^{p \times n}$ with $d \in \mathbf{R}^p$, $p \leq n$, consider the constrained version of problem (2), given by

$$\begin{aligned} &\text{minimize} \quad \left\| \tilde{A}x - \tilde{b} \right\|_2^2 \\ &\text{subject to} \quad Cx = d. \end{aligned} \quad (3)$$

Write the KKT linear system that would be used to solve problem (3) in terms of A, b, λ, C, d .

- (d) (10 points) Consider a mass at rest in 1-dimension (initial position and velocity being 0) on a frictionless surface, with $m = 1/2$ with a 2-vector of forces $f = (f_1, f_2)$ applied for one second each. Applying Newton's laws gives the final velocity and position after two seconds:

$$\begin{aligned} v^{\text{fin}} &= 2f_1 + 2f_2 \\ p^{\text{fin}} &= 3f_1 + f_2. \end{aligned} \quad (4)$$

We would like to bring the mass to final velocity $v^{\text{fin}} = 0$ and penalize the squared distance of the final position p^{fin} to a reference point p^{ref} . To prevent the force from being too large, we add a penalty on the squared 2-norm of f . Written concretely, we aim to solve the following optimization problem:

$$\begin{aligned} &\text{minimize} \quad \left\| p^{\text{fin}} - p^{\text{ref}} \right\|_2^2 + \lambda \|f\|_2^2 \\ &\text{subject to} \quad v^{\text{fin}} = 0. \end{aligned} \quad (5)$$

Solve problem (5) with $p^{\text{ref}} = 1$, $\lambda = 1$. Give the optimal force vector f^* . What is the final position of the mass?

Hint: You can reuse your results from part (c).

Solution.

(a) Choose

$$\tilde{A} = \begin{bmatrix} A \\ \sqrt{\lambda}I \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

(b) The normal equations tell us

$$x^\star = (\tilde{A}^T \tilde{A})^{-1} (\tilde{A}^T \tilde{b})$$

To simplify, recognize that:

$$\tilde{A}^T \tilde{A} = \begin{bmatrix} A^T & \sqrt{\lambda}I \end{bmatrix} \begin{bmatrix} A \\ \sqrt{\lambda}I \end{bmatrix} = A^T A + \lambda I, \quad \tilde{A}^T \tilde{b} = \begin{bmatrix} A^T & \sqrt{\lambda}I \end{bmatrix} \begin{bmatrix} b \\ 0 \end{bmatrix} = A^T b.$$

So, the normal equations simplify to

$$x^\star = (A^T A + \lambda I)^{-1} A^T b.$$

We do not need any assumptions on A since $\tilde{A}$ always has linearly independent columns due to the stacking with I .

(c) The KKT conditions are given by:

$$\begin{bmatrix} 2\tilde{A}^T \tilde{A} & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ z^\star \end{bmatrix} = \begin{bmatrix} 2\tilde{A}^T \tilde{b} \\ d \end{bmatrix},$$

which simplifies to

$$\begin{bmatrix} 2(A^T A + \lambda I) & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ z^\star \end{bmatrix} = \begin{bmatrix} 2A^T b \\ d \end{bmatrix}. \quad (6)$$

(d) In terms of the physics equations and given constants, problem (5) can be written as

$$\begin{aligned} & \text{minimize} \quad \left\| \begin{bmatrix} 3 & 1 \end{bmatrix} f - 1 \right\|_2^2 + \|f\|_2^2 \\ & \text{subject to} \quad \begin{bmatrix} 2 & 2 \end{bmatrix} f = 0. \end{aligned}$$

That is, we have a multiobjective constrained least squares problem with

$$A = \begin{bmatrix} 3 & 1 \end{bmatrix}, \quad b = 1, \quad C = \begin{bmatrix} 2 & 2 \end{bmatrix}, \quad d = 0.$$

If we substitute these values into (6), we get:

$$\begin{bmatrix} 20 & 6 & 2 \\ 6 & 4 & 2 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 0 \end{bmatrix}$$

Writing this out gives the system:

$$20f_1 + 6f_2 + 2z = 6$$

$$6f_1 + 4f_2 + 2z = 2$$

$$2f_1 + 2f_2 = 0.$$

The last equation implies $f_1 = -f_2$, which substituted back in gives

$$14f_1 + 2z = 6$$

$$2f_1 + 2z = 2.$$

Solving the last system and substituting gives the solution $(f_1^*, f_2^*, z^*) = (1/3, -1/3, 2/3)$, which gives $p^{\text{fin}} = 2/3$.

2. Problem 2 (35 points). Found money in a dumpster.

You find \$50,000 in a dumpster behind Sherrerd Hall and you decide to invest all of it and *not care about the risk*. You would like to invest at least \$10,000 in municipal bonds yielding 4% returns, and at most \$20,000 in a high risk, high yield fund returning 10%. Your goal is to maximize your average portfolio return after one year. And let's hope that you make a lot of money before whoever owns the dumpster-money comes to find you.

Use the following information in setting up your problem. Let w_1 be the fraction of the money you put into municipal bonds, and w_2 be the fractional amount you invest in your high yield fund. Therefore, $w = (w_1, w_2)$ gives the fraction of your portfolio held in each asset class, with $\mathbf{1}^T w = 1$. You allow only long positions. Vector $\mu = (0.04, 0.1)$ is the vector of average historical returns for each asset.

- (a) (15 points) Set up a linear programming problem whose solution will maximize your average portfolio return.
- (b) (10 points) Solve the linear program you set up in part (a) graphically. That is, draw your constraints, identify the feasible set, draw contour lines parallel to your objective function, and use that graph to identify the optimal solution $w^* = (w_1^*, w_2^*)$.
- (c) (10 points) Recall that the standard form of an LP is given by

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0 \end{array}$$

Write the linear program you formulated in part (a) in standard form and explicitly state the corresponding values of c , x , A , and b .

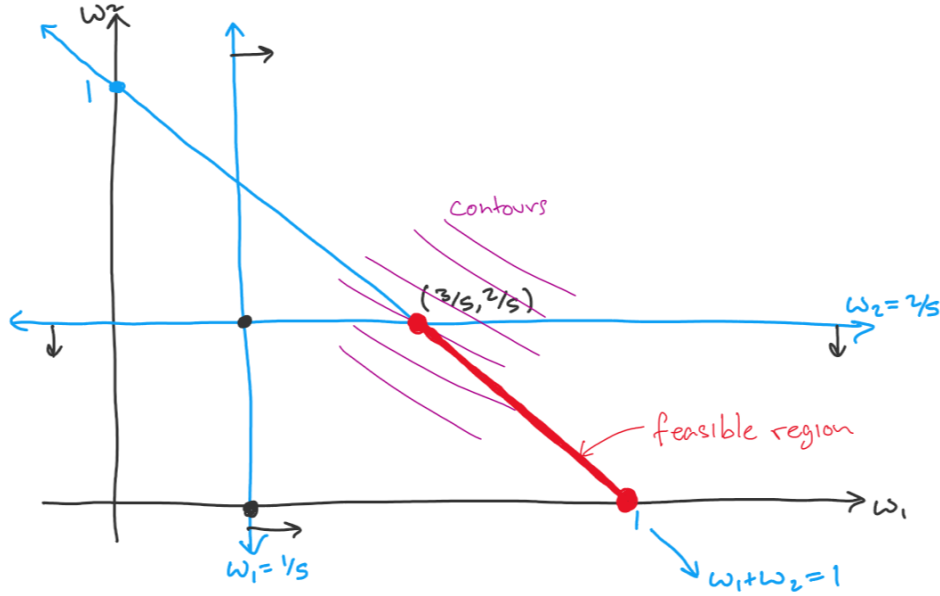
Solution.

- (a) Our objective is to maximize the average returns $\mu^T w$, where

$$\mu = \begin{bmatrix} 0.04 \\ 0.1 \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \text{with } w_1 \geq \frac{10,000}{50,000} = \frac{1}{5}, \text{ and } w_2 \leq \frac{20,000}{50,000} = \frac{2}{5}$$

Therefore, our LP is:

$$\begin{array}{ll} \text{maximize} & \mu^T w \\ \text{subject to} & \mathbf{1}^T w = 1 \\ & w_1 \geq 1/5 \\ & w_2 \leq 2/5 \\ & w \geq 0 \end{array} \quad \xrightarrow{\text{explicitly}} \quad \begin{array}{ll} \text{maximize} & 0.04w_1 + 0.1w_2 \\ \text{subject to} & w_1 + w_2 = 1 \\ & w_1 \geq 1/5 \\ & w_2 \leq 2/5 \\ & w \geq 0 \end{array}$$



(b) Our analysis by graphing is below:

Since our feasible set is on the line $w_1 + w_2 = 1$, there are only two points to consider.

w_1	w_2	$\mu^T w$
1	0	0.04
3/5	2/5	0.064 *

From the table, we can see that $(w_1^*, w_2^*) = (3/5, 2/5)$.

Alternatively, we can note that since investments in w_2 will elicit higher returns, our solution would benefit from $w_2^* \neq 0$. Every dollar moved from w_1 to w_2 receives a higher average return rate, and so $(w_1^*, w_2^*) = (3/5, 2/5)$.

(c) To reformulate our LP into standard form, we negate the objective and include slack variables $s_1, s_2 \geq 0$. We obtain,

$$\begin{aligned}
 &\text{minimize} && -0.04w_1 - 0.1w_2 + 0s_1 + 0s_2 \\
 &\text{subject to} && w_1 + w_2 = 1 \\
 &&& w_1 - s_1 = 1/5 \\
 &&& w_2 + s_2 = 2/5 \\
 &&& w, s \geq 0
 \end{aligned}$$

So that,

$$c = \begin{bmatrix} -0.04 \\ -0.1 \\ 0 \\ 0 \end{bmatrix}, \quad x = \begin{bmatrix} w_1 \\ w_2 \\ s_1 \\ s_2 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1/5 \\ 2/5 \end{bmatrix}$$

3. *Problem 3 (30 points). Medical treatment.*

You work for a Orfizer Therapeutics, where you are tasked with spearheading the development of a new drug. You need to decide how much to include from an array of n ingredients. This drug will be given to m patients. Let x_i be the amount of the i th ingredient you decide to add, and $x \in \mathbf{R}^n$ the vector of the amounts for every ingredient. You cannot put a negative amount of an ingredient in the recipe. For each patient $k \in \{1, 2, \dots, m\}$, the symptoms $y_k \in \mathbf{R}^d$, are a linear function of the active ingredients, *i.e.*,

$$y_k = A_k x,$$

where $A_k \in \mathbf{R}^{d \times n}$. The elements $(A_k)_{ji}$ represent how an active ingredient i affects symptom j . Your goal is to design a drug such that the symptoms y_k of patient k are as close as possible to their desired values y_k^{des} , for $k = 1, \dots, m$.

- (a) (10 points) Formulate the problem of minimizing the sum, across all patients, of the maximum absolute deviation between their symptoms and the target values. Which type of optimization problem do you obtain? Explain why.
- (b) (10 points) The management tells you that each unit of active ingredient i costs $c_i > 0$ for $i = 1, \dots, n$, and that you should minimize the expenses as well. You decide to add the cost of the active ingredients to the objective function. Formulate this new optimization problem *both* with (i) general vector norms, and (ii) as a linear program.
- (c) (10 points) Your drug seems to be working really well, and Orfizer would like to apply for FDA approval. In order to do so, you have very stringent requirements: the maximum absolute deviations between the symptoms of each patient and the target must be less than a federally approved threshold $\epsilon > 0$. Formulate this problem as a linear program, by adapting the problem in part (b.ii) in order to take into account this requirement, while minimizing the active ingredient costs.

Solution.

- (a) For patient k , the maximum absolute deviation in symptoms is given by

$$\|y_k - y_k^{\text{des}}\|_{\infty} = \|A_k x - y_k^{\text{des}}\|_{\infty}.$$

By summing across all of the patients and adding the nonnegativity constraints on the ingredients, we arrive at the following optimization problem:

$$\begin{aligned} & \text{minimize} && \sum_{k=1}^m \|A_k x - y_k^{\text{des}}\|_{\infty} \\ & \text{subject to} && x \geq 0. \end{aligned} \tag{1}$$

Problem (1) is a *piecewise linear* optimization problem due to the inclusion of the ∞ norms.

- (b) (i) The cost of the active ingredients is given by $c^T x$ so the new optimization problem can be written as

$$\begin{aligned} & \text{minimize} && c^T x + \sum_{k=1}^m \|A_k x - y_k^{\text{des}}\|_{\infty} \\ & \text{subject to} && x \geq 0. \end{aligned} \tag{2}$$

- (ii) To write (2) as an LP, we need to remove the ∞ norms. Since there are m of them, we introduce a new m -vector variable t and bound each component separately. This gives the following LP:

$$\begin{aligned} & \text{minimize} && c^T x + \mathbf{1}^T t \\ & \text{subject to} && A_k x - y_k^{\text{des}} \leq t_k \mathbf{1}, \quad k = 1, \dots, m \\ & && -(A_k x - y_k^{\text{des}}) \leq t_k \mathbf{1}, \quad k = 1, \dots, m \\ & && x \geq 0. \end{aligned} \tag{3}$$

- (c) The difference for this problem is that we replace each auxiliary variable t_k by ϵ and remove the $\mathbf{1}^T t$ from the objective since the FDA requirement is strict and we cannot optimize for it. This gives the following LP:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && A_k x - y_k^{\text{des}} \leq \epsilon \mathbf{1}, \quad k = 1, \dots, m \\ & && -(A_k x - y_k^{\text{des}}) \leq \epsilon \mathbf{1}, \quad k = 1, \dots, m \\ & && x \geq 0. \end{aligned} \tag{4}$$